AI & SOCIETY (2026) 41:1737–1748
https://doi.org/10.1007/s00146-025-02488-3

RESEARCH

The construction of the role of AI in qualitative data analysis in the social sciences

Trena Paulus[1] · Jessica Nina Lester[2] · Charlie Davis[1]

Received: 7 April 2025 / Accepted: 3 July 2025 / Published online: 14 July 2025

Abstract

Generative AI (AI) is being promoted for and adopted by social science researchers at a rapid pace. This shift has enormous implications for research outcomes. A better understanding of the language that authors are using to construct a version of social science research processes that incorporate AI can inform reporting guidelines and best practices. Qualitative data analysis is a particularly important area of focus as it has historically positioned humans *as* the research instrument. Human researchers are coming from positions in the world that always impact their understanding of the data. In this paper, we used discourse analysis methods to interrogate how researchers publishing in peer-reviewed social science journals are proposing and/or explaining their use of generative AI to conduct qualitative data analysis. We gathered a corpus of 29 articles through extensive database and direct tables of content searches and documented the journal discipline, paper purpose, and AI platforms. We noted five discursive stances: (1) qualitative data analysis is inherently problematic; (2) qualitative data analysis is easily, and will inevitably be, automated; (3) AI will disrupt methods without (immediately) replacing humans; (4) AI-human hybrid methods are the future; and (5) de-centering ethical concerns and model limitations. We conclude by suggesting how these "AI as qualitative data analyst" discourses will complicate the relationship between qualitative methods and humans as researchers.

Keywords Generative artificial intelligence · ChatGPT · Qualitative data analysis · Qualitative research · Discourse analysis

1 Introduction

The role of generative artificial intelligence (AI) in scientific research has increasingly become a normative practice, widely promoted by scholars at the highest levels (Wang 2023). While non-generative, or discriminative, machine learning AI systems have been around for years, generative AI in the form of OpenAI's ChatGPT first came into mainstream public awareness in late 2022. From a research perspective, it is particularly important to note that these AI systems are not evaluating the 'new information' they generate against any validated base of knowledge. Rather, they are designed to ensure that their outputs match the structure of their training data input. As Thornton (2023) described, "for a machine learning model, an output does not have to *be* correct, it must *appear* correct" (p. 27), leading to a plausibility bias. In 2024, the U.S. National Academies of Sciences, Engineering, and Medicine hosted a workshop with a stated goal to: "explore the future of AI in terms of its role as an autonomous researcher performing scientific discovery" (National Academies of Science, Engineering, Medicine 2024). Similarly, the Nobel Turing Challenge seeks to develop a "highly autonomous AI and robotics system that can make major scientific discoveries" (Nobel Turing Challenge 2024). Beyond academia, technology companies have situated AI as science's savior (Schmidt 2023).

While the promise of AI excites many, researchers are grappling with the visible and hidden consequences of its use. Scholars across a range of disciplines have taken up a variety of viewpoints—from Ethan Mollick's (2024) positioning of AI as a 'co-intelligence' to Bender and Hanna's (2025) consistent pushback on 'AI as scientist', distributed

✉ Trena Paulus
paulust@etsu.edu

Jessica Nina Lester
jnlester@iu.edu

Charlie Davis
davisck@etsu.edu

[1] East Tennessee State University, Johnson City, USA

[2] Indiana University Bloomington, Bloomington, USA

through their podcast, *Mystery AI Hype Theater 3000*. Papers have reported on scholar (Fecher et al. 2023) and student (Alshwiah 2024) perspectives on research process impact, as well as encouraging its embrace in academic life (Lin 2023).

Such conversations have been taking place across social science fields, including business and management (Mariani and Dwivedi 2024), marketing (Hao and Liu 2024), economics (Korinek 2023), psychology (Chenneville et al. 2024; Schueller and Morris 2023), archaeology (Cobb 2023), political studies (Finn et al. 2024), library science (Hosseini and Holmes 2023), and statistics (Goodale 2024). For the purpose of this paper, we are contrasting the social sciences (the study of humans, communities and social systems) with the natural sciences (study of the physical world). Scholars have begun to focus on how AI will impact specific parts of the social science research process, such as the literature review (Atkinson 2023); data collection methods, including developing surveys (Olivos and Liu 2024), interview protocols (Parker et al. 2023), and focus groups (Anis and Olisa 2024); transcription (Taylor 2024); academic writing (Pigg 2024); and manuscript preparation (Chemaya and Martin 2024).

As phases of the research process are conducted alongside, or handed over to, AI, the threat of 'deepfakes' (a product that seems real but was generated by AI) is becoming apparent (Silver 2024). That is, not only can AI-generated photographs, videos, artwork, and text be taken to represent true events and human points of view, but entire research papers can now be generated without human intervention. Sison et al. (2024) argued that "the main danger ChatGPT presents is the propensity to be used as a 'weapon of mass deception'" (p. 4843). The authors specifically described this in terms of the possibility of unsafe, unreliable and untrustworthy ChatGPT output being used for deception in academic work, disinformation campaigns, and criminal activities. Arora et al. (2024) discussed what they called the 'AI-human hybrid' in market research, while Lily et al. (2023) considered whether and how society is perceiving ChatGPT to have human-like traits. Perhaps most significantly for qualitative social scientists are papers discussing whether AI can serve as a proxy for human research participants (Argyle et al. 2023; Burleigh and Wilson 2024; Fostier et al. 2024).

Qualitative social science research methodologists and researchers, too, have been considering how AI will impact their work (Christou 2023). Qualitative approaches focus on understanding human sense-making in context, giving particular attention to how people navigate social worlds (Paulus and Lester 2023). These methodological approaches have historically assumed that humans are the primary research instrument, fully engaged in data collection, analysis, and interpretation. Some approaches to analysis focus on categorical thinking: engaging in summarizing, categorizing, and identifying general patterns in a given dataset (Freeman 2016), all of which are strengths of AI (van Manen 2023). AI tools have been part of qualitative data analysis (QDA) software platforms for many years, as keyword extraction, sentiment analysis, and topic modeling. Less than a year after the release of ChatGPT in November of 2022, ATLAS.ti and MAXQDA integrated OpenAI large language models into their platforms, with the CAQDAS Networking Project at the University of Surrey hosting webinars and workshops for the community to discuss potential impacts and consequences (e.g. Silver 2023).

There are, however, many different ways by which qualitative data is analyzed and interpreted, and this variation can get lost in discussions around the impact of generative AI on qualitative methods. As senior scholars and practitioners of a range of qualitative research methodologies, we believe that the consequences for using or not using AI are significant for research workflows and knowledge outcomes. Our stance on generative AI has been in constant evolution since ChatGPT was introduced in the fall of 2023. We have been committed to staying informed about the training practices and foundations of large language models (Mollick 2024; Bender and Hanna 2025), raising awareness of the power structures behind its development and adoption (Hao 2025), and, simultaneously, engaging with early adopters in the qualitative research community who are attempting to demonstrate its value for our work. Our collaborative work as authors has consistently explored the intersection of technological innovation and qualitative inquiry with a critical, reflexive stance.

Given the emergent and flexible nature of qualitative research, as well as the vast landscape of methods and methodologies, we highlight the necessity of scholars engaging in regular reflexivity when working at the intersections of research and any new technologies (Paulus and Lester 2023). However, to date, scholarly work that examines the intersection of technology and qualitative research typically flattens this relationship. For example, Woods et al. (2016) noted that researchers who discussed their use of NVivo or ATLAS.ti qualitative data analysis (QDA) software in their research reports typically did so by simply naming their methodology as 'generic' qualitative research and justified their use of technology with claims that it would improve the rigor of the study. There is no such thing as 'generic' qualitative research, and simply using software does not, of course, improve the quality of a study (Paulus and Lester 2022). As a result of these findings, guidelines for reporting the use of QDA software in more robust and meaningful ways have been offered (Paulus et al. 2017).

As a contribution to the special issue entitled, AI Technology as Human Interaction, in this paper we explore the discursive framing of AI systems. We explore how a very human-centered endeavor, making meaning of human social life via qualitative inquiry, is intersecting with AI systems that promise a new approach to data analysis. We focus specifically on how researchers are working up new discourses of what it means to do QDA in the age of generative AI—potentially flattening what it has to offer. The interpretative nature of many qualitative research designs (such as ethnography, narrative, discourse analysis, phenomenology) has typically included positioning the human researcher *as* the primary instrument (e.g., Watt 2007); that is, researchers have been understood to be, rather than neutral or objective actors, entering into a given study from a particular position in the world. This subjective positionality necessarily impacts their understanding (and analysis) of data, primarily in critical, positive ways. As such, it has been of particular interest to note how AI is being positioned as a co-researcher (Mollick 2024). We report findings from our discourse analysis of peer-reviewed journal articles, asking: How do authors describe the relationship between QDA and generative AI?

2 Discourse analysis and AI use

Discourse analysis is a well-established method for the study of language as social practice (Woods and Kroger 2000) and has been previously used to analyze research publications (e.g., Paulus et al. 2017; Woods et al. 2016). While still a relatively small body of literature, previous AI discourse analysis studies informed our own work. Avnoon and Oliver (2023) noted what they termed the 'rhetorical dance' that radiologists performed around how AI would disrupt and/or be adopted into their work. These discourses included integration through cooperation and co-producing with AI; absorbing AI as another useful technology but subordinating it to the human radiologists; fighting off the AI threat by undermining its legitimacy and strengthening boundaries; and, finally, full assimilation of radiology into the larger field of AI. Within the context of higher education, McGrath et al. (2024) drew upon Bearman et al. (2022) to explore the discourses present in empirical studies of generative AI. They noted how Bearman et al.'s (2022) discourses of imperative response; altering authority; and the 'dystopia is now, utopia is just around the corner' duality were present throughout the research literature. The discourse of imperative response was characterized as: "AI is a force of unprecedented and inevitable change that requires higher education to either resist or adapt" (McGrath et al. 2024, p. 4). The discourse of altering authority highlighted how AI is challenging the locus of authority and agency throughout the university and "depicts shifting power dynamics between humans and machines and actors such as teachers, students, corporations and university" (McGrath et al. 2024, p. 4). In their analysis, AI was depicted as posing a challenge to the role and purpose of both teachers and students.

Similarly, Hammond et al. (2023) identified three primary discourses used on automated-paraphrasing websites targeting students. First, they identified the sheep discourse, characterized as "disguised messaging appealing to innocent students" (p. 9). This discourse functioned to position the paraphrasing site as a helpful resource that is low cost and easy to use. The second discourse ("dark wolf") was described as constructing "…an easy world of speed and efficiency while the student escapes the frustration of learning and avoid[s] the risk of negative emotion associated with being caught by plagiarism detection tools" (p. 10). Finally, acceptability and normalization discourses of 'everyone does it' were also prevalent. The authors noted that the gray wolf discourse blurred the sheep and wolf discourses by problematizing both writing and learning itself. It functioned to frame normal learning as requiring effort and as a negative experience to be avoided.

Finally, Munk et al. (2024) examined what evoking algorithms is 'doing' in scientific literature, focusing on how algorithms are often positioned as having agency. They noted that the 'expert' scientific discourse charges algorithms only with the capacity to *solve* problems. Algorithms are rarely blamed for *causing* problems; and, if problems arise from algorithms, it is up to another algorithm to solve them.

Paulus and Marone (2025) analyzed how QDA software company websites described the impact of AI, identifying four discursive dilemmas: (a) data analysis constructed as automated insight-generation versus systematic meaning-making; (b) data analysis transformed into chatting with documents versus analyzing data; (c) data analysis characterized as happening at high speed versus with high engagement; and (d) the novelty of AI-driven analysis and outcomes versus retention of human agency. These dilemmas posed challenges to the essence of what it means to analyze data from various epistemological stances. Building on Paulus and Marone (2025), we analyzed, in this study, the emerging discourses around AI and QDA in peer-reviewed journal articles.

3 Method

We gathered a corpus of peer-reviewed journal articles proposing or reporting the use of generative AI for QDA. As qualitative researchers immersed in these discourses, this seemed a logical starting point given the previously discussed literature. To identify relevant articles, we conducted both database keyword searches and direct tables of contents searches, adapting procedures outlined by Paulus et al. (2024) and Woods et al. (2016). Our search for relevant articles occurred in several phases. First, we conducted searches in EBSCO, ProQuest and Web of Science using the keywords "generative AI", "ChatGPT", "large language models" and "artificial intelligence" in combination with "qualitative data analysis." Second, we created keyword search alerts in Google Scholar and Semantic Scholar. Finally, keyword searches were conducted in the tables of content of 67 journals that publish qualitative research. We started with the journals reviewed by Paulus et al. (2024) and added other social science and applied discipline-specific journals known to publish qualitative work. We limited the search to peer-reviewed journals published in English in 2023 and 2024, accounting for the release date of ChatGPT. Peer review remains the gold standard of publication quality, and English is the dominant language of our research team. While we manually reviewed hundreds of initial hits, only 29 articles were relevant to the purpose of our review.

We began the analysis by documenting the journal discipline, purpose of the paper, and AI platform. Next, we used discourse analysis (Wood and Kroger 2000) to make sense of how the authors constructed versions of QDA and the relationship between the human researcher and generative AI platforms. Discourse analysis rests on the assumption that language choices perform actions (functioning to construct a particular relationship between the research process and AI) rather than represent a static reality (representing an objective truth about this relationship). This methodological approach is grounded in the perspective that the human experience of the world is discursively and socially constructed. Thus, the research team analyzed the language used in the articles, both individually and collaboratively, by engaging with the following questions:

1. How is the QDA process characterized, described and constructed?
2. How is the role of the human characterized, described and constructed?
3. How is the role of AI characterized, described and constructed?
4. How is the relationship between humans and AI characterized, described and constructed?

By closely analyzing language use, we noted how discourses and metaphors around technology function in particular ways, such as attributing agency to the generative AI, rather than the researcher. We engaged in repeated readings, making analytic and theoretical memos around what language choices were doing, constructing, and describing. We met together to compare, contrast, and systematize our interpretations. Following Wood and Kroger (2000), strategies and procedures included the analysis of text structure and organization (i.e., what is there and what is not, and how it is structured) and examination of multiple functions of discourse segments (i.e., what the language choices are doing and how). We organized our findings around broad patterns, illustrating what these patterns accomplish and paying particular attention to the potential consequences that these discursive patterns might have for consumers of research.

4 Findings

Table 1 provides an overview of the journal disciplines, paper purpose, and AI platforms. Several articles fell into more than one category. Most of the articles were published in methodology journals, which is unsurprising given that researchers are still in the early adoption or pre-adoption phase. Some scholars published conceptual pieces speculating on AI use alongside ethical and other methodological considerations. Articles comparing human analysis with AI analysis were the most common paper purpose, followed by workflow papers describing in detail how AI was, or could be, used for data analysis. Only three studies were found to report that AI had been used to conduct the study (Hajra

Table 1 Article categorization by journal discipline, purpose, and AI platform

Journal disciplines	Purpose of the paper *some papers represented more than one purpose	AI platforms *most papers discussed multiple types of AI
Qualitative methodology/evaluation (10)	Comparison (11)	ChatGPT (18)
Health (5)	Workflow (9)	Generative AI (10)
Social media analysis (4)	Conceptual (8)	Qualitative data analysis software integration (5)
Education/social work/human services (4)	Research reports (3)	ChatPDF (1)
Information systems (2)		Bard (1)
Tourism (2)		
Business (1)		
Ethics (1)		

2023; Malik et al. 2024; Zhang and Tandoc 2024). As adoption continues, we anticipate more studies will fall into this category. Finally, ChatGPT was the most common AI platform, followed by generically described generative AI, with five papers discussing the integration of AI into existing QDA platforms such as ATLAS.ti and MAXQDA.

Broadly, we identified five discursive patterns used to construct the relationship between humans, AI and qualitative data analysis. These were: (1) qualitative data analysis is inherently problematic; (2) qualitative data analysis is easily, and will inevitably be, automated; (3) AI will disrupt methods without (immediately) replacing humans; (4) AI-human hybrid methods are the future, and (5) de-centering ethical concerns and model limitations. The italicized text is used to highlight discursive moves that illustrate our arguments most clearly.

1. "Accelerating a process that is exhausting and tiring": Qualitative data analysis is inherently problematic

The article introductions often positioned generative AI as able to overcome characteristics of qualitative methods that were framed as inherent *weaknesses*. In our experience as scholars, qualitative research designs are often selected because they allow for a richer, more detailed picture of the environment being studied. Deeper understandings result from extended time in the field and close engagement with participants and locations. Framing qualitative research as *slow, labor intensive, exhausting*, and *expensive* functioned to position its strengths as de facto problems, which opened the door to the adoption of AI as a solution. Claiming that qualitative research is *subjective* and *biased* also provided such an opening, when, in actuality, the underlying epistemological foundations of qualitative research designs typically do not make any claims to objectivity. Tai et al. (2024), for example, described qualitative work as comprised of "meticulous, *time-consuming*, and *labor-intensive* manual coding" (p. 1) which was positioned as relying on *subjective* processes. They also described 'positionality' as a feature that "introduces *bias* into the analysis and interpretation" (p. 1). By constructing qualitative inquiry as "time-consuming", "labor intensive", and "subjective", Tai et al. (2024) implicitly positioned these qualities as *problematic* versus *central* core strengths of choosing a qualitative design.

Like Tai et al. (2024), Carius and Teixeira (2024) wrote that, "the use of Artificial Intelligence in the coding process is a way of *accelerating* a process that is, to a certain extent, *exhausting* and *tiring*, leaving the researcher with *little time* to dedicate to the subliminal aspects of the material and, therefore, make important inferences regarding the proposed material" (p. 11). AI was described by Chubb (2023) as being able to *expedite* and *reduce human labor* and load. Similarly, Victor et al. (2023) connected the integration of large language models into ATLAS.ti with an increase in analytic *efficiency*, reducing the "*time* and *effort*" of the researcher engaged in analysis (p. 566). Dahal (2024) further noted that AI can "*overcome data scarcity* and delve deeper into research questions…allowing for more nuanced analyses and broader understanding… it can *save time and effort* when human resources are scarce…" (p. 725–726). Here, the small datasets often intentionally used in qualitative studies are positioned as a "data scarcity" problem that AI can solve.

Broadly, many of the articles, particularly those within the health sciences, forwarded what is perhaps best characterized as a *moral imperative* to do important work *faster*. As a possible acknowledgement that qualitative research is typically designed to be slow, scholars from the health sciences fields made the argument that, in order to solve real-world problems in a timely manner, it is the researchers' *obligation* to use AI. Parker et al (2023) noted that there are large amounts of unstructured data in health systems that remain unanalyzed due to *time* and *cost*. If these piles of data are left unanalyzed it creates missed opportunities to do the right thing, they argued, therefore AI is a solution. Relatedly, Prescott et al. (2024) acknowledged that qualitative methods are beneficial, but lamented that they take time and ultimately delay generating knowledge that is needed to solve health problems.

A few scholars, such as Hitch (2024), acknowledged and rejected the assumption that such essential characteristics of qualitative research are inherently problematic. While noting some potential benefits of AI, such as systematicity, providing an alternate perspective, and identifying patterns not initially noticed by humans, Hitch also argued that: "This potential benefit... should not be mistaken as a means to *increase 'objectivity'*" (p. 601). Hitch acknowledged that AI has been "proposed as a way of *validating* qualitative analysis which reflects the *detached* perspectives of the *neo-positivist paradigm*... [which may] adopt this more readily due to this perceived alignment to their underlying values" (p. 601). Hitch recognized that efficiency is a contentious concept—choosing efficiency over being consistent with the underlying epistemological assumptions of a study, even if it requires more time, are two different 'solutions'. Still, Hitch framed *time as a luxury and privilege* and argued that AI could help researchers work *efficiently* through large datasets, writing: "chat GPT could complete a first pass of data in matter of minutes" (p. 602), which would ensure that *humans* could use their time creatively *deepening insights and doing nuanced interpretation*. Anis and French (2023) also positioned AI as opening up ways for qualitative researchers to spend time doing activities that mattered, such as thinking, deliberating, and developing interpretations. That is, AI was constructed as an entity that could free humans from more time-consuming, and perhaps less rewarding, parts of analysis.

As a logical extension of this argument, we noted that humans were described at times in these articles as not even really *wanting* to do key parts of the tiresome and time-consuming work required by QDA. This orientation thus opened up space for AI to be positioned as releasing humans to do more important things with their limited time. Extensive time spent coding, analyzing, and discussing analysis with others were treated as tasks that humans would, really, just prefer not to do. Walsh and Pallas-Brink (2023) described and resisted this discourse well: "In applying AI to qualitative data, the implied problem is that qual is not quant. As ethnographers, we know qual is not quant for a very good reason—qualitative data offers insights centered on 'how' and 'why,' versus the 'what' and 'how many' of quantitative" (p. 548). Moreover, some articles positioned human researchers as problematic—unable to achieve consistent and cohesive coding, for example. Humans were described as envying the superior abilities of AI (Carius and Teixeira 2024). These superior AI abilities, as Christou (2024) proposed, included improved speed and scalability to manage large datasets, automate, group similar items, summarize, find relationships, recognize patterns, offer alternative perspectives, and much more.

2. "Themes inferred by the model": Qualitative data analysis is easily, and will inevitably be, automated.[1]

The articles we analyzed were situated along a spectrum of epistemological orientations, though most took up post-positivist perspectives, foregrounding methods such as thematic analysis, grounded theory, and content analysis (see Chubb 2023, for an exception). These methods were generally positioned as involving coding and categorizing. Analysis was framed as amenable to automation, bolstering the argument that AI integration could increase efficiency. None of the articles we reviewed analyzed images or videos, nor did they use discursive, posthuman, or creative analysis methods. This may, in part, reflect limitations in our search terms, which did not explicitly include visual or image-generating AI tools such as DALL-E.

Since QDA was positioned as a relatively straightforward process of summarizing and identifying patterns, scholars had the grounding to construct studies comparing human analysis with AI analysis through a 'media comparison' (see Buchner and Kerres 2023, for a discussion of the limitations of media comparison study designs). There was a general assumption that AI *should* be able to do what humans do, perhaps better. Eleven of the 29 papers we analyzed included AI-human comparisons. For example, Spangler et al. (2024) mused that while non-qualitative data analysis includes:

> mathematical computations [which] have a *logical* beginning and reach a *regimented* end . . . on the other hand, qualitative studies are *very reliant on a human factor*, the researcher, to determine when all possible data has been exhausted and identified and then to determine how to give meaning to what is obtained… *AI analyzing data should come to a relatively same conclusion analyzing the same data as a pair of experienced researchers* (p. 323).

In these comparative approaches, some authors began with an AI analysis and then determined whether the human analysis achieved the same outcome, whereas others approached the comparison in reverse.

The goal of these studies was typically to assess whether and to what extent human and AI analysis aligned. We noted that authors acknowledged yet also minimized any inconsistences. For example, De Paoli (2024) attempted to partially reproduce six phases of thematic analysis with ChatGPT. De Paoli (2024) concluded that "the model can infer *most* of the main themes from previous research…this shows that using LLMs [large language models] to perform an inductive [thematic analysis] is *viable* and offers a *good degree* of validity" (page 1). De Paoli went on to say: "…There is *no denying* that the model can *infer* codes and themes and even identify patterns which *researchers did not consider relevant* and *contribute to better qualitative analysis*" (p. 18). Here, rather than treating the inconsistency as a model limitation, it is assumed that the AI-identified codes and themes that the humans *did not consider relevant* should be given equal credibility and authority as study findings.

The assumption that there will and should be a role for AI in the qualitative data analysis process was prevalent across the papers. ChatGPT was positioned as having advantages for working with large datasets, with scholars recognizing "the importance of *combining AI tools with human expertise* to refine and interpret the results *accurately*" (Hajra 2023, p. 182). There was a tone of *inevitability* in the discourses evoked. De Paoli (2023), as one example, concluded with: "in the end it does seem *inevitable* that qualitative researchers, especially in the social sciences, *will have to engage* with these models in ways that *can* help do their work better" (p. 20). Similarly, Hamilton et al. (2023) described ChatGPT as "a powerful tool to *supplement complex human-centered* tasks... such tools *will* become an additional tool to facilitate research tasks" (p. 1).

Even when AI was acknowledged to have limitations, the majority of the authors positioned AI as inevitably useful. For example:

[1] While our analysis only focused on how scholars wrote about the use of generative AI for qualitative data analysis, some of the included articles, such as Dahal (2024), Head et al. (2023), Marshall and Naff (2024), and Victor et al. (2023), did in fact discuss potential uses of AI for other phases of the research process.

> Despite ChatGPT's commendable performance, limitations include issues around contextual understanding, bias, and researcher agency constraints . . . this study *acknowledges these limitations*, emphasizing their recognition as vital for comprehensive interpretation. ChatGPT, combined with manual analysis, *enhances overall robustness, despite potential challenges* with interpretive aspects . . . (Hajra 2023, p. 182).

As another example, Dahal (2024), after noting some limitations claimed that "*nevertheless*... AI technologies can enhance researchers' capabilities, enabling them to become more resourceful and self-reliant in their roles" (p. 727). Scholars took up a type of fatalistic discourse – AI will be adopted, so researchers should all get on board. Hitch (2024) acknowledged that it "is not whether it *should* be adopted, but *how to* best adopt it. The *revolution* is here, and it has the *potential to significantly enhance* and *transform* the work of qualitative researchers from multiple traditions and approaches" (p. 604).

Van Manen (2023), one of the few scholars to push back on AI adoption, cautioned that ChatGPT "cannot really thematize, let alone theorize, in a meaningfulness sense. Instead, [it] offers an illusion of meaningful reading" (p. 1136). He noted that in methodological traditions like phenomenology the focus is on understanding human *experience*, something ChatGPT cannot offer. Similarly, Davison et al (2024) suggested that it is the "researcher's creative and conceptual ability to discern meaning, salience and interconnectedness of logic in emerging themes" and that AI cannot "capture the nuances of the research environment, body language, facial expressions, tone of voice, interactions between the researcher and the research subjects, and the researcher's own accumulated understanding of the domain." (p. 4). These less frequently articulated viewpoints will be revisited in the following section on the discourses taken up around the role of humans in data analysis.

3. "Challenging conventional distinctions": AI will disrupt methods without (immediately) replacing humans

Alongside the framing of data analysis as a problem that is easily solvable with AI, scholars also took up a discourse of disruption-but-not-replacement. AI was constructed as being both disruptive (i.e., altering how qualitative data analysis is done) and disrupting (i.e., resulting in innovations that will likely create new problems). Victor et al. (2023) positioned generative AI as having "the potential to *reshape many core activities* in qualitative research... the integration of ChatGPT within ATLAS.ti will certainly increase qualitative coding *speed*, but the integration of this generative AI technology *might diminish the quality and trustworthiness* of analyses" (p. 566–568). Similar to Victor et al., Morgan (2023) asked whether using ChatGPT "might *disrupt* coding of data segments as the *dominant paradigm*"…noting that, "if one believes the publicity about AI-based programs, they offer the *possibility* of *simply* 'chatting' with your data *to understand its complexities*, rather than *going through the detailed coding processes* that currently dominate qualitative data analysis" (p. 1). Morgan further mused that "only time will tell what direction AI-based analysis will take, but given the *disruptive power* of ChatGPT, that time may be quite short" (p. 9). The disruptive capacities of AI were framed as potentially replacing a mainstay practice, coding. A contrast was made between coding as something that analysts must endure ("go through") versus the, presumably more enjoyable, 'simply chatting' with data as mediated through AI systems.

Parker et al. (2023) argued that researchers should use AI to ensure that they will "be on the *cutting edge of emerging technology* which is *revolutionizing* many fields" (p. 5). Törnberg (2024) noted that generative AI has "capacities that have been traditionally *understood to be distinctly human*" (p. 1) and that large language models "appear to have *granted machines access to the realm of ideas, values and beliefs about the world* within which texts are situated, *historically thought to be an exclusively human* domain…" (p. 13). He argued that AI can "draw on interpretive skills and can make inferences based on contextual understanding…" requiring "*new ways of thinking* of computational text annotation, one that is more akin to human text coding…" (p. 3). Törnberg went on to claim that these models are "*challenging conventional distinctions* within the social sciences: between qualitative and quantitative studies, and between close and distant reading... [opening] new forms of large-scale inductive interpretive research, representing new methodological terrain" (p. 12). Indeed, he argued, radical scientific advances will come from these new tools, "*triggering a paradigm shift*" (p. 12).

Alongside the "AI as disruptor/disruption" discourse were reassuring claims that there would still be a role for humans. There was an effort to display that humans and AI platforms were compatible and should work together, rather than considering AI to be a *replacement* for humans. For example, Spangler et al. (2024) concluded that an AI bot was a good starting point for thematic analysis, but humans were still needed for *nuanced* analysis and *emotional* understanding. Chubb (2023) characterized AI as a useful *research assistant but not as good as a human*, noting that the human must still review AI products. That this move essentially creates more work that humans never asked for was not addressed by any of the authors. Instead, AI was frequently referenced as "simply a tool"; one that could be used *by* the human, but that would not *take over the analysis* (Zhang and Tandoc 2024). In this way, the indispensable human position was maintained, albeit decentered as the only, or even primary, authority. AI was centered as the primary, initial actor that needed humans to "check" it for accuracy (e.g. Hayden 2023). In this *human as validation agent* positioning, some of the authors were explicit in pointing to the current impossibility of replacing humans due to some of AI's weaknesses, such as hallucinations and the generation of false information (Marshall and Naff 2024; Van Manen 2023). This positioning of AI as inevitable and essential but also ultimately unreliable poses a dilemma for scholars.

A minority of articles acknowledged that the unique aspects of qualitative inquiry (i.e., engaging in reflexivity; taking up innovative and creative analytic methods) were beyond the abilities of current AI systems; thereby requiring human involvement. Van Manen (2023) noted:

> ChatGPT does not 'put something into words' in a meaning-oriented manner. Instead, it 'merely' generates bare texts based on a complex predictive model of what words are likely to follow from the preceding words, phrases, sentences, and so forth while also shaped by the prompts entered by the user (p. 1136).

Further, Wachinger et al. (2024) discussed how AI cannot engage in reflexivity, and as such humans still must do this as it is fundamental to the epistemologies underlying many qualitative inquiry methodologies.

Authors noted that while, for now, humans remain indispensable, this would not always be the case. For example, Hitch (2024) noted that "AI can *augment* but not *replace* the work of human researchers" (p. 596), but that the "the boundary between human and non-human interpretations of data is likely to become *blurrier* over time" (p. 602).

4. "A virtual colleague": Hybrid methods as the future

Given that, despite its limitations, AI is being positioned as both inevitable and disruptive to QDA, and that, at the same time, humans will not (yet) be replaced, the question becomes: what will the future bring? Across many of the articles, new, hybrid methods of analysis were proposed and positioned as imminent. Parker et al. (2023) argued:

> *combining* a qualitative researcher with [generative AI] could be deployed in many settings, *reducing time and costs* and improving context…until LLMs are more prevalent, a *human* interaction can help translate patient experience by *contextualizing* data rich in social determinant indicators which may otherwise go *unanalyzed* (p. 1).

Prescott et al. (2024) concluded that "*consistency in the themes generated by human coders and GenAI* suggests that these technologies *hold promise in reducing the resource intensiveness* of qualitative thematic analysis; however, the *relatively lower reliability* in coding between them suggests that *hybrid approaches are necessary*" (p. 1). They went on to note that humans are better at identifying *nuanced* and interpretive themes, and thus *collaboration will improve efficiency* as well as *mitigate potential ethical risks posed by the AI*. Prescott et al. provided two paths—using AI for analysis first, after which humans would review the outcomes, or, in contrast, humans would begin the analysis, followed by AI, after which researchers would reflect on any discrepancies.

This new hybrid research endeavor included a variety of roles for AI. Hamilton et al. (2023) predicted AI would become an additional *tool* in the researcher toolbelt, alongside Google Scholar, citation management platforms and grammar-checking software. Dahal (2024) described AI as a *co-author*, a *conversational platform* and a *research assistant* that could both enhance and hinder research. Hitch (2024) exclaimed, "much to my surprise, I found myself referring to ChatGPT as '*they/them*' in conversations with humans... I think of it as a *virtual colleague*, a *novice* that can make a useful contribution" (p. 604). Mazeikiene and Kasperiuniene (2024) suggested that AI could be a *co-analyst, consultant* and *trainer*, and centered the role of humans in decision making and interpretation.

Some proposed entirely new AI-integrated research methodologies, rather than just re-envisioning the human and AI role relationship. Sun et al. (2025) positioned ChatGPT as its own unique research method alongside grounded theory methodology: "While *ChatGPT and Grounded Theory operate differently* due to their fundamental nature, there are several *parallels in the processes* from initial coding, developing, refining, and interconnecting categories, to theoretical integration, validation, and refinement" (p. 4). Head et al. (2023) proposed an AI-generated workflow for evaluation research: "Evaluation processes can be supported by LLMs across the *following evaluation workflow*: (1) summarizing existing text, (2) comparing multiple texts, (3) analyzing individual texts, (4) analyzing tables and images, and (5) creating (generating) text" (p. 37–38). Christou (2024) outlined *thematic analysis through AI* as a new method. Tai et al. (2024) proposed "a *systematic methodology for using LLMs as a qualitative research tool* that can aid in the deductive coding of interview data" (p. 1). Mazeikiene and Kasperiuniene (2024) described a "research design for *AI-enhanced qualitative analysis*" (p. 2507), framing this as a methodological approach.

Mazeikiene and Kasperiuniene (2024) suggested that "our efforts to extract an explanation of the analysis of the data through various prompts *has not been successful*…. ChatGPT doesn't reveal actual algorithms in conversation thus the application of algorithms and generation of outputs *remains obscure* for its users…." (p. 2515). Lack of success and methodological obscurity should sound alarming to researchers, but such limitations did not undermine this proposed new hybrid methodology. In fact, the authors argued that "GAI is successful in analyzing initial raw data, open coding and elevating codes to a higher degree of abstraction and revealing main patterns in the data… delineat[ing] the efficiency and effectiveness of GenAI in data analyzing" (p. 2521).

Broadly, then, the majority of the articles positioned humans as necessary and central to this new future hybrid approach to data analysis. Yet, while humans were constructed as *central*, they were no longer positioned as the *primary* research instrument, as has typically been the case in qualitative traditions. Rather, the human role was relegated to reviewing the automated, fast and efficient AI output. While some scholars, such as Davison et al (2024) and van Manen (2023), kept humans at the core of the research process—resisting claims that ease and speed were more important than ethical and high-quality analysis—these were minority voices.

5. "Humans are also black-boxed and subjective": Decentering ethical concerns and model limitations

Most authors delayed, until the end of the articles, discussions of ethical concerns, model limitations or other critiques of AI use in qualitative analysis (see Van Manen 2023, as an exception). Concerns that were discussed included the frequency of errors ('hallucinations'), data privacy risks, ambiguous data ownership, interpretative insufficiency, and biased outcomes. Concerns were not typically positioned as a cause for serious alarm or reasons to delay adoption; rather, they were simply stated as temporary issues that were likely to be resolved in the near future and were articulated in a backgrounded, minimized way.

Törnberg (2024), for instance, mentioned ethics, legality and reproducibility as challenges to AI-generated outcomes, but countered any arguments against its use by reminding us that "many of the same criticisms can also be *leveraged against human* coders—who, after all, are similarly *black-boxed [in their thinking] and subjective*" (p. 12). Similarly, Victor et al. (2023) noted that "researchers should confirm compliance with ethical standards, including data privacy and security, to ensure the research meets all ethical requirements" (p. 573), making any AI-related problems the responsibility of the individual researcher, rather than the technology companies, to solve. Morgan (2023) acknowledged the potential for algorithmic bias, AI-generated hallucinations, and ethical problems, particularly around using research data to train LLMs. However, Morgan reassured the reader that his own study encountered no problems with hallucinations, with this formulation serving to minimize any significant concerns. While the negative environmental impacts caused by the natural resources needed to sustain server farms that support generative AI technologies were noted by Head et al. (2023), we found no authors emphasizing this. Nor did authors mention the poorly treated data workers or the unauthorized use of intellectual property as training data foundational to generative AI. Marshall and Naff (2024) did note that LLM training data disproportionately represents White and middle to high SES people—so AI should be understood as "culturally misaligned with participants from minoritized, marginalized, and underrepresented backgrounds, which is particularly dangerous if AI qualitative analyses begin to be interpreted as 'truth'" (p. 94).

Prescott et al. (2024) perhaps best illustrated this discursive pattern, wherein they described limitations at the very end of the article, mentioning data privacy, algorithmic bias, inequality, fact fabrication, plagiarism, breaches of data privacy, and informed consent challenges around adopting AI for research. Despite these grave limitations and ethical concerns, the authors suggested that "future researchers should continue to prudently investigate and monitor both the potential benefits and risks associated with groundbreaking technologies such as GenAI when incorporating them into the scientific process with data from vulnerable populations" (p. 9). The authors did not advise against the use of this 'groundbreaking' technology for social science research. Rather, it is left up to individual researchers to investigate and monitor technological developments, while also determining and following best practices for their use, even when such practices have not yet fully been articulated.

5 Conclusions and limitations

Similar to other discourse studies of AI, our analysis found that the majority of authors are engaging in what Avnoon and Oliver (2023) termed the rhetorical dance. This dance is one in which researchers recognize that AI will most certainly disrupt, if not wholeheartedly transform, the social science landscape regardless of potential threats. McGrath et al. (2024) noted that within higher education more broadly, AI is being positioned in the discourse as an *inevitable* change. We found across the papers that AI was constructed as something that there is simply no way around. Like Hammond et al.'s (2023) study of how automated paraphrasing sites positioned their utility for students, we, too, found discourses highlighting the "ease and efficiency" it will bring to the qualitative data analysis process. This construction points to the realities of living and working within a context driven by neoliberal and capitalistic goals (e.g., Olssen and Peters 2005). That is, the mandate to produce more, faster, is part of the framework within which the majority of social science researchers find themselves working.

In *Nature*, Messeri and Crockett (2024) warned of a world in which "proposed AI solutions" to conducting research will make us "vulnerable to illusions of understanding in which we believe we understand more about the world than we actually do" (p. 49). They argued that the four visions of AI for scientific research that have been proposed to span the research cycle (AI as oracle that can wade through the vast quantities of scholarly literature; as surrogate silicon research subjects and data generators; as quants that can automate data preparation and analysis; and as arbiters to serve as proxy editors and peer reviewers) are *not* inevitable, and certainly pose epistemic risks, such as production of a scientific monoculture. Offering a similar warning, Heersmink et al. (2024) suggested that anthropomorphizing large language models can lead to unwarranted trust toward their output, given that the "data and algorithmic opacity and phenomenological and informational transparency can make it difficult for users to calibrate their trust appropriately" (p. 40).

We, therefore, are troubled by the AI-human relationship that is being constructed around qualitative data analysis. Rather than "*problematically* time intensive", we argue that the time-intensive nature of qualitative work is not, in fact, troubling. Rather, the time-intensive nature of qualitative data analysis is often part of what is needed to generate a deep and detailed understanding of a human phenomenon. Long-term engagement with data is one way by which we—as inquirers—can ensure trustworthiness and authenticity of claims (Tracy 2010). Moreover, qualitative data analysis has long been inherently shaped by the human researchers driving the process. By design, the researcher is indeed central to the interpretive process. This reality is neither a weakness nor a problem; rather, it is a core feature of qualitative research that moves analysts to commit to reflexivity. While indeed many qualitative researchers are already claiming that AI applications are useful and result in more efficient research practices, we call for caution and hesitation in their adoption. What is offered to us currently are applications that have not offered theoretically and methodologically sound pathways toward conducting qualitative data analysis. Moreover, these applications by and large focus on categorizing and thematizing data—which aligns with only a small number of qualitative analytic approaches (e.g., content analysis, thematic analysis, etc.). In fact, the dangers of engaging with these applications to conduct analysis are not yet fully known.

Davison et al. (2024) suggested that guidelines for AI use need to be flexible, evolving along with the technologies. Indeed, the use of AI in academic research and publishing has resulted in increasing calls for guidelines to ensure ethical research practice and academic integrity. In fact, a call for clear guidelines around both the use of AI in research and the reporting of its use is echoed by multiple authors who argue that the lack of comprehensive guidelines can lead to confusion among researchers, reviewers, and editors (e.g., Ganjavi 2024; Zhong 2023). Several major research organizations (e.g. the American Psychological Association and the Committee on Publication Ethics) have already developed journal policies outlining when and how to disclose the use of generative AI. We recognize that the 'inevitability' discourses of AI use will likely continue, making it critical to develop ways to reflexively acknowledge the consequences.

Given the shifting landscape around appropriate uses of generative AI and social science research, scholars may be using AI for various research tasks without reporting it. Our analysis highlighted a striking trend: social science researchers are increasingly willing – perhaps even eager – to delegate some aspects of their interpretive authority to AI technologies. This widespread and growing practice is perhaps being fueled by neoliberal narratives that prioritize efficiency, speed, and the illusion that such tools lead researchers to more objective outcomes. Yet, as qualitative methodologists and researchers, we are compelled to ask: at what cost do we embrace AI in our interpretive work?

Author contributions TP, JNL and CD conducted the data collection and analysis and wrote up the manuscript. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

References

Alshwiah A (2024) Students' perceptions of an artificially intelligent chatbot as a support tool to develop their research skills. Qual Res J. https://doi.org/10.1108/QRJ-06-2024-0117

Anis S, French JA (2023) Efficient, explicatory, and equitable: why qualitative researchers should embrace ai, but cautiously. Bus Soc 62(6):1139–1144. https://doi.org/10.1177/00076503231163286

Anis AM, Olisa N (2024) The end of traditional focus groups? Scaling up qualitative research quick, yet maintaining depth. New Trends Qual Res 20(1):1–18. https://doi.org/10.36367/ntqr.20.1.2024.e799

Argyle LP, Busby EC, Fulda N, Gubler JR, Rytting C, Wingate D (2023) Out of one, many: using language models to simulate human samples. Polit Anal 31:337–351. https://doi.org/10.1017/pan.2023.2

Arora N, Chakraborty I, Nishimura Y (2024) EXPRESS: AI-human hybrids for marketing research: leveraging LLMs as collaborators. J Mark. https://doi.org/10.1177/00222429241276529

Atkinson CF (2023) Cheap, quick, and rigorous: artificial intelligence and the systematic literature review. Soc Sci Comput Rev. https://doi.org/10.1177/08944393231196281

Avnoon N, Oliver AL (2023) Nothing new under the sun: Medical professional maintenance in the face of artificial intelligence's disruption. Big Data Soc. https://doi.org/10.1177/20539517231210269

Bearman M, Ryan J, Ajjawi R (2022) Discourses of artificial intelligence in higher education: a critical literature review. Higher Educ 1:17. https://doi.org/10.1007/s10734-022-00937-2

Bender EM, Hanna A (2025) The AI con: how to fight big tech's hype and create the future we want. Harper, New York

Buchner J, Kerres M (2023) Media comparison studies dominate comparative research on augmented reality in education. Comput Educ 195:104711. https://doi.org/10.1016/j.compedu.2022.104711

Burleigh C, Wilson AM (2024) Generative AI: Is authentic qualitative research data collection possible? J Educ Technol Syst. https://doi.org/10.1177/00472395241270278

Carius AC, Teixeira AJ (2024) Artificial Intelligence and content analysis: the large language models (LLMs) and the automatized categorization. AI Soc. https://doi.org/10.1007/s001

Chemaya N, Martin D (2024) Perceptions and detection of AI use in manuscript preparation for academic journals. PLoS ONE. https://doi.org/10.1371/journal.pone.0304807

Chenneville T, Duncan B, Silva G (2024) More questions than answers: ethical considerations at the intersection of psychology and generative artificial intelligence. Transl Issues Psychol Sci 10(2):162–178. https://doi.org/10.1037/tps0000400

Christou P (2023) How to use artificial intelligence (AI) as a resource, methodological and analysis tool in qualitative research? Qual Report 28(7):1968–1980. https://doi.org/10.46743/2160-3715/2023.6406

Christou P (2024) Thematic analysis through artificial intelligence (AI). Qual Report 29(2):560–576. https://doi.org/10.46743/2160-3715/2024.7046

Chubb LA (2023) Me and the machines: possibilities and pitfalls of using artificial intelligence for qualitative data analysis. Int J Qual Methods 22:1–16. https://doi.org/10.1177/16094069231193593

Cobb PJ (2023) Large language models and generative AI, oh my! Adv Archaeol Pract 11(3):363–369. https://doi.org/10.1017/aap.2023.20

Dahal N (2024) How can generative AI enhance or hinder qualitative studies? A critical appraisal from South Asia, Nepal. Qual Report 29(3):722–733. https://doi.org/10.46743/2160-3715/2024.6637

Davison RM, Chughtai H, Nielsen P, Marabelli M, Iannacci F, Van Offenbeek M, Tarafdar M, Trenz M, Techatassanasoontorn AA, Díaz Andrade A, Panteli N (2024) The ethics of using generative AI for qualitative data analysis. Inf Syst J 34(5):1433–1439. https://doi.org/10.1111/isj.12504

De Paoli S (2024) Performing an inductive thematic analysis of semi-structured interviews with a large language model: an exploration and provocation on the limits of the approach. Soc Sci Comput Rev 42(4):997–1019. https://doi.org/10.1177/08944393231220483

Fecher B, Hebing M, Laufer M, Pohle J, Sofsky F (2023) Friend or foe? Exploring the implications of large language models on the science system. AI Soc. https://doi.org/10.1007/s00146-023-01791-1

Finn P, Bell LC, Tatum A, Leicht CV (2024) Assessing ChatGPT as a tool for research on US state and territory politics. Polit Stud Rev. https://doi.org/10.1177/14789299241268652

Fostier J, Leemans E, Meeussen L, Wulleman A, Doren SV, Coninck DD, Toelen J (2024) Dialogues with AI: comparing ChatGPT, Bard, and human participants' responses in in-depth interviews on adolescent health care. Future 2:30–45. https://doi.org/10.3390/future2010003

Freeman M (2016) Modes of thinking for qualitative data analysis. Routledge

Ganjavi C (2024) Publishers' and journals' instructions to authors on use of generative artificial intelligence in academic and scientific publishing: bibliometric analysis. BMJ. https://doi.org/10.1136/bmj-2023-077192

Goodale T (2024) Using generative AI to help with statistical test selection and analysis. MSOR Connect 22(3):10–16

Hajra V (2023) Unveiling the aesthetics of sacred elephant interactions: a ChatGPT-driven analysis of Tripadvisor reviews. J Qual Res Tourism 4(2):175–183

Hamilton L, Elliott D, Quick A, Smith S, Choplin V (2023) Exploring the use of AI in qualitative analysis: a comparative study of guaranteed income data. Int J Qual Methods 22:1–13

Hammond KM, Lucas P, Hassouna A, Brown S (2023) A wolf in sheep's clothing? Critical discourse analysis of five online automated paraphrasing sites. J Univ Teach Learn Pract 20(7):8. https://doi.org/10.53761/1.20.7.08

Hao K (2025) Empire of AI. Penguin Random House, Westminster

Hao A, Liu H (2024) Guest editorial: artificial intelligence application and future research directions in interactive marketing. J Res Interact Mark 18(1):1–5. https://doi.org/10.1108/JRIM-07-2023-0231

Hayden D (2023) Generative AI: mining housing data with a higher-powered shovel. Cityscape A J Policy Dev Res 25(3):199–206. https://doi.org/10.2307/48749677

Head CB, Jasper P, McConnachie M, Raftree L, Higdon G (2023) Large language model applications for evaluation: Opportunities and ethical implications. New Direct Eval 178–179:33–46. https://doi.org/10.1002/ev.20556

Heersmink R, de Roolj B, Vazquez JC, M., & Colombo, M. (2024) A phenomenology and epistemology of large language models: Transparency, trust, and trustworthiness. Ethics Inf Technol 26:41. https://doi.org/10.1007/s10676-024-09777-3

Hitch D (2024) Artificial intelligence augmented qualitative analysis: The way of the future? Qual Health Res 34(7):595–606. https://doi.org/10.1177/10497323231217392

Hosseini M, Holmes K (2023) The evolution of library workplaces and workflows via generative AI. Coll Res Libr 84(6):836–842. https://doi.org/10.5860/crl.84.6.836

Korinek A (2023) Generative AI for economic research: use cases and implications for economists. J Econ Liter 61(4):1281–1317. https://doi.org/10.1257/jel.20231736

Lily AEA, Ismail AF, Abunsar FM, Al-Lami F, Abdullatif AKA (2023) ChatGPT and the rise of semi-humans. Human Soc Sci Commun 10(626):1–12. https://doi.org/10.1057/s41599-023-02154-3

Lin Z (2023) Why and how to embrace AI such as ChatGPT in your academic life. R Soc Open Sci 10(230658):1–11. https://doi.org/10.1098/rsos.230658

Malik M, Youk S, Weber R (2024) Beyond the screen: exploring moral understanding via user comments on YouTube short films. J Media Psychol 36(4):231–243. https://doi.org/10.1027/1864-1105/a000430

Mariani M, Dwivedi YK (2024) Generative artificial intelligence in innovation management: a preview of future research developments. J Bus Res 175:114542. https://doi.org/10.1016/j.jbusres.2024.114542

Marshall DT, Naff DB (2024) The ethics of using artificial intelligence in qualitative research. J Empir Res Hum Res Ethics 19(3):92–102. https://doi.org/10.1177/15562646241262659

Mazeikiene N, Kasperiuniene J (2024) AI-enhanced qualitative research: insights from Adele Clarke's situational analysis of TED Talks. Qual Report 29(9):2502–2526. https://doi.org/10.46743/2160-3715/2024.7652

McGrath C, Farazouli A, Cerratto-Pargman T (2024) Generative AI chatbots in higher education: a review of an emerging research area. High Educ. https://doi.org/10.1007/s10734-024-01288-w

Messeri L, Crockett MJ (2024) Artificial intelligence and illusions of understanding in scientific research. Nature 627(7):49–58. https://doi.org/10.1038/s41586-024-07146-0

Mollick E (2024) Co-intelligence. Penguin Random House, Westminster

Morgan DL (2023) Exploring the use of artificial intelligence for qualitative data analysis: the case of ChatGPT. Int J Qual Methods 22:1–10. https://doi.org/10.1177/16094069231211248

Munk AK, Jacomy M, Ficozzi M, Jensen TE (2014) Beyond artificial intelligence controversies: What are algorithms doing in the

scientific literature? Big Data Soc. https://doi.org/10.1177/20539517241255107
Nobel Turning Challenge (Retrieved on December 4, 2024). News. Available at: https://www.nobelturingchallenge.org/
National Academies of Sciences, Engineering, Medicine (Retrieved December 4, 2024). AI for Scientific Discovery - A Workshop. Available at: https://www.nationalacademies.org/event/40455_10-2023_ai-for-scientific-discovery-a-workshop
Olivos F, Liu M (2024) ChatGPTest: Opportunities and cautionary tales of utilizing ai for questionnaire pretesting. Field Methods 1:14. https://doi.org/10.1177/1525822X241280574
Olssen M, Peters MA (2005) Neoliberalism, higher education and the knowledge economy: from the free market to knowledge capitalism. J Educ Policy 20(3):313–345
Parker RD, Mancini K, Abram MD (2023) Natural language processing enhanced qualitative methods: an opportunity to improve health outcomes. Int J Qual Methods 22:1–6. https://doi.org/10.1177/16094069231214144
Paulus T, Lester J (2022) Doing qualitative research in a digital world. Sage, Thousand Oaks, CA
Paulus T, Lester J (2023) Digital qualitative research workflows: A reflexivity framework for technological consequences. Int J Soc Res Methodol. https://doi.org/10.1080/13645579.2023.2237359
Paulus TM, Marone V (2025) "In minutes instead of weeks": discourse dilemmas on AI for qualitative data analysis. Qual Inq 31(5):395–402
Paulus T, Woods M, Atkins D, Macklin R (2017) The discourse of qualitative data analysis software (QDAS): reporting practices of ATLAS.ti and NVivo users with implications for best practices. Int J Soc Res Methodol 20(1):35–47
Paulus T, Pope L, Bower K (2024) Ways that qualitative researchers engage in 'technological reflexivity': a meta-synthesis. Qual Inquiry 31:305–318
Pigg S (2024) Research writing with ChatGPT: a descriptive embodied practice framework. Comput Compos 71:102830. https://doi.org/10.1016/j.compcom.2024.102830
Prescott MR, Yeager S, Ham L, Rivera Saldana CD, Serrano V, Narez J, Paltin D, Delgado J, Moore DJ, Montoya J (2024) Comparing the efficacy and efficiency of human and generative AI: qualitative thematic analyses. JMIR AI 3:e54482. https://doi.org/10.2196/54482
Schmidt E (2023) This is how AI will transform the way science gets done. Available at: https://www.technologyreview.com/2023/07/05/1075865/eric-schmidt-ai-will-transform-science/
Schueller SM, Morris RR (2023) Clinical science and practice in the age of large language models and generative artificial intelligence. J Consult Clin Psychol 91(10):559–561. https://doi.org/10.1037/ccp0000848
Silver C (2023) Qualitative-AI: Hoo-ha or Step Change (Christina Silver). CAQDAS Webinar 041. Available at: https://www.youtube.com/watch?v=SplxzzhxbjU
Silver C (2024) The risk of qualitative deepfake is a reality. Available at: https://www.qdas.co.uk/post/ai-deepfake-qualitative-analysis
Sison AJG, Daza MT, Gozalo-Brizuela R, Garrido-Merchan E (2024) ChatGPT: more than a "weapon of mass deception" ethical challenges and responses from the human-centered artificial intelligence (HCAI) perspective. Int J Hum Comput Interact 40(17):4853–4872. https://doi.org/10.1080/10447318.2023.2225931
Spangler S, Beasley S, Lingelbach K (2024) Analyzing artificial intelligence frameworks: phenomenologists against machines. Issues Inf Syst 25(4):332–332. https://doi.org/10.48009/4_iis_2024_126
Sun H, Kim M, Kim S, Choi L (2025) A methodological exploration of generative artificial intelligence (AI) for efficient qualitative analysis on hotel guests' delightful experiences. Int J Hosp Manag 124:103974. https://doi.org/10.1016/j.ijhm.2024.103974
Tai RH, Bentley LR, Xia X, Sitt JM, Fankhauser SC, Chicas-Mosier AM, Monteith BG (2024) An examination of the use of large language models to aid analysis of textual data. Int J Qual Methods 23:1014. https://doi.org/10.1177/16094069241231168
Taylor Z (2024) Using chat GPT to clean qualitative interview transcriptions: a usability and feasibility analysis. Am J Qual Res 8(2):153–160. https://doi.org/10.29333/ajqr/14487
Thornton I (2023) A special delivery by a fork: Where does artificial intelligence come from? New Direct Eval 2023:23–32
Törnberg P (2024) Large language models outperform expert coders and supervised classifiers at annotating political social media messages. Soc Sci Comput Rev. https://doi.org/10.1177/08944393241286471
Tracy SJ (2010) Qualitative quality: eight 'big-tent' criteria for excellent qualitative research. Qual Inq 16(10):837–851
Van Manen M (2023) What does ChatGPT mean for *Qualitative Health Research* ? Qual Health Res 33(13):1135–1139. https://doi.org/10.1177/10497323231210816
Victor BG, Sokol RL, Goldkind L, Perron BE (2023) Recommendations for social work researchers and journal editors on the use of generative AI and large language models. J Soc Soc Work Res 14(3):563–577. https://doi.org/10.1086/726021
Wachinger J, Bärnighausen K, Schäfer LN, Scott K, McMahon SA (2024) Prompts, pearls, imperfections: comparing ChatGPT and a human researcher in qualitative data analysis. Qual Health Res. https://doi.org/10.1177/10497323241244669
Walsh S, Pallas-Brink J (2023) The ethnographer in the machine. In: Ethnographic praxis in industry conference proceedings, pp 538–554. https://www.epicpeople.org/ethnographer-in-the-machine-everyday-experiences-with-AI-enabled-data-analysis
Wang H (2023) Scientific discovery in the age of artificial intelligence. Nature 620(3):47–60. https://doi.org/10.1038/s41586-023-06221-2
Watt D (2007) On becoming a qualitative researcher: the value of reflexivity. Qual Report 12(1):82–101. https://doi.org/10.46743/2160-3715/2007.1645
Wood LA, Kroger RO (2000) Doing discourse analysis: Methods for studying action in talk and text. Sage Publications, London. https://doi.org/10.4135/9781452233291
Woods M, Paulus T, Atkins D, Macklin R (2016) Advancing qualitative research using qualitative data analysis software? Reviewing potential vs practice in published studies using ATLAS.ti and NVivo, 1994–2013. Soc Sci Comput Rev 34(5):597–617
Zhang L, Tandoc EC (2024) Hashtagging for inclusion: Complex identities in Singaporean gay men's social representation on Instagram. Soc Med Soc 10(3):1–12. https://doi.org/10.1177/20563051241269299
Zhong J (2023) The endorsement of general and artificial intelligence reporting guidelines in radiological journals: a meta-research study. BMC Med Res Methodol 23:1. https://doi.org/10.1186/s12874-023-02117-x